

Sensor Timing Firmware

The Teensy 4.1 firmware separates physical data acquisition from timing I/O. The Linux host continues acquiring all four cameras, both VLP-16s, and the Xsens MTi-30 continuously. The firmware can qualify a common reference edge, schedule future trigger pulses, observe per-camera exposure and IMU sync edges, and publish timing diagnostics. It neither transports payload data nor proves that a host driver accepted a hardware trigger.

All new timing outputs are disabled and unassigned in [firmware_config.h](#). A build becomes actuating only after the corresponding pins, polarities, line drivers, configuration, and wiring-verification gates are supplied. The scheduler then requires stable reference periods before running, stops on a missing or out-of-range reference, rejects late trigger deadlines, requires camera ExposureActive feedback, and latches faults rather than continuing open-loop. Relative timing remains unqualified until the assembled board is measured with an oscilloscope or logic analyzer. The DS3231 INT/SQW output is open-drain; the assembled board must pull it up to 3.3 V or use a verified 3.3 V level shifter. A module's 5 V pull-up must never reach the non-5-V-tolerant Teensy. The currently available square wave is not a verified UTC/GNSS phase source. Published firmware time references use a monotonic epoch beginning at the first qualified edge and explicitly do not claim UTC. On the existing `sensor_msgs/TimeReference` diagnostics, `header.stamp` is only the ROS publication-receipt time and `time_ref` is the unmapped relative MCU epoch. These topics are non-authoritative and are not a ROS-time calibration contract.

Electrical boundary

The installed-camera inventory must be checked against the physical labels. The current Teledyne reference names the official color model FG-PGE-50S5C-IP; the repository's FG-PGE-50S5C-C-IP spelling is not assumed to be a second electrical variant.

For the current Forge IP67 family, the M8 eight-pin GPIO interface assigns pin 1 to 8-24 V camera input power, pin 2/Line0 to the opto-isolated input, pin 3 to opto ground, pin 6/Line1 to the opto-isolated output, and pin 7 to camera power ground. Published typical thresholds are 1.55 V low and 2.6 V high for Line0; Line1 is specified as approximately 0.56 V low and 5 V high under the vendor's measurement conditions. The Teensy accepts only 0-3.3 V inputs and its normal output recommendation is 4 mA. Therefore neither Forge line connects directly to the MCU. The board must provide characterized, 3.3 V-safe buffered or isolated inputs and outputs, defined returns, and protected/fused 8-24 V camera branches when GPIO power is used. Use IP-rated M8 cables or sealing plugs to retain the camera's ingress rating.

Firmware provides both four independently configurable logical outputs and one common hardware-fanout output, but sequential GPIO writes are not claimed to be simultaneous. Select and characterize the common fanout or an output latch when all four Line0 edges must be simultaneous. Configure each camera separately for `FrameStart`, the selected Line0 rising edge, and `Line1 ExposureActive`; both device-side functions have separate false-by-default firmware gates. Verify every accepted trigger and exposure edge on the assembled harness before enabling the firmware gate. The current host profile remains free-running and unchanged.

The VLP-16s acquire continuously. Their input is a clock/phase reference, not a scan trigger. The official manual requires a PPS high above 3.0 V and below 5.0 V, low below 1.2 V, and at least 2 mA source current. NMEA and PPS must alternate; PPS-to-NMEA separation is at least 50 ms and the NMEA sentence must finish at least 300 ms before the next PPS. The current firmware has no verified UTC/GNSS source, so it does not emit the old placeholder GPRMC sentence. Dual-VLP phase locking additionally requires a locked PPS, an RPM multiple of 60 between 300 and 1200, and distinct measured phase offsets configured and saved on the sensors. The MCU only distributes the shared edge. Its two GPIO writes are sequential; use and characterize a common electrical fanout when the two LiDAR edges must share one physical transition.

The encased MTi-30 synchronization contract accepts 0-0.8 V low and 2.5-20 V high on `SyncIn/ClockSync`; `SyncOut` is 0-0.4 V low and above 2.9 V high. Firmware cannot determine the selected MTi `SyncSettings`. The optional MCU output is only a time-event marker for the continuously sampled IMU; it must not select `StartSampling` or reduce navigation data to the camera's 10 Hz cadence. It therefore remains disabled until polarity, voltage, cable pin, the `SyncIn` event, the `SyncOut` feedback event, and a corresponding packet marker are independently verified in MT Manager.

Firmware structure and validation

`sensor_sync.h` owns the side-effect-free state machine, bounded edge mailboxes, and continuous rollover-safe relative-epoch mapper. GPIO ISRs only deposit timestamped edges; the timer ISR is the sole owner of scheduler, feedback, fault, and relative-time state. Mailbox overflow latches synchronization off. `sensor_sync_runtime.h` owns Teensy GPIO, camera exposure midpoint capture, IMU events, feedback deadlines, and safe inactive outputs.

`firmware_pin_contract.h` rejects cross-owner aliases, RTC-I2C pin collisions, and unsupported telescope PWM/ADC assignments before either runtime initializes. `main.ino` owns ROS publication, RTC edge labelling, the timer ISR, and the existing disabled telescope runtime.

The pure scheduler is suitable for non-actuating host compilation and state transition checks. These checks are performed from an ephemeral harness rather than a repository validator. They check logic only. A valid Teensy build requires a Teensyduino core, roserial headers, and RTCLib. Acceptance additionally requires a current-limited bench supply, no motor or sensor payloads connected for initial bring-up, safe inactive outputs during boot/reset/disconnect, instrumented edge-to-edge skew and jitter, deliberate missing-reference and feedback-fault injection, MTi packet-marker verification, VLP PPS-lock/phase-lock status, and camera exposure/frame-counter correlation. Passing simulation or host tests is not physical synchronization evidence.

The telescope motor driver must also have external hard pulldowns and a hardware driver-disable path so reset, boot, disconnected MCU, or high-impedance GPIO cannot energize either half-bridge. Firmware fail-closed states supplement that electrical interlock; they do not replace it.

Primary references

- Teledyne FLIR, [Forge 5 MP GigE camera family](#), model and acquisition-feature reference.
- Teledyne FLIR, [Forge IP67 GPIO electrical characteristics](#), M8 pinout, thresholds, delays, and line ratings.
- Teledyne FLIR, [Forge IP67 acquisition control](#), FrameStart trigger-source/mode/overlap contract.
- Ouster (Velodyne), [VLP-16 User Manual, 63-9243 Rev. F](#), PPS/NMEA electrical, timing, and phase-lock requirements.
- Analog Devices, [DS3231 datasheet](#), open-drain INT/SQW electrical contract.
- Xsens, [MTi 10-series and MTi 100-series User Manual, MT0605P](#), encased MTi connector, SyncIn/SyncOut, and ClockSync electrical characteristics.
- PJRC, [Teensy 4.1 technical specifications](#), MCU GPIO voltage and current limits.